

```
0 1 10 10
suhani@siri:~/Hospital-Patient-Scheduler$ cd /home/suhani/Hospital-Patient-Scheduler
suhani@siri:~/Hospital-Patient-Scheduler$ gcc src/main.c src/scheduler.c src/fileio.c src/metrics.c
suhani@siri:~/Hospital-Patient-Scheduler$ ./a.out
```

HOSPITAL PATIENT SCHEDULER

1. Non-Preemptive Priority
2. Preemptive Priority
3. Compare Both
4. Exit

Enter choice: 2

PROCESS INPUT

Enter number of processes (1-100): 3

----- Process 1 -----

Enter Process ID: 2
Enter Arrival Time: 2
Enter Burst Time: 4
Enter Priority: 3

----- Process 2 -----

Enter Process ID: 5
Enter Arrival Time: 1
Enter Burst Time: 8
Enter Priority: 6

----- Process 3 -----

Enter Process ID: 3
Enter Arrival Time: 5
Enter Burst Time: 7
Enter Priority: 3

Aging Configuration

1. Disable Aging
2. Enable Aging

Enter choice: 1

Save results to output file?

1. No
2. Yes

Enter choice: 1

PREEMPTIVE PRIORITY RESULTS

Process States

2 : Terminated
5 : Terminated
3 : Terminated

Execution Order

IDLE -> 5 -> 2 -> 3 -> 5

Gantt Chart

IDLE	5	2	3	5	
0	1	2	6	13	20

Process Results

ID	Arrival	Burst	Priority	Completion	Turnaround	Waiting	Response
2	2	4	3	6	4	0	0
5	1	8	6	20	19	11	0
3	5	7	3	13	8	1	1

Average Waiting Time : 4.00
Average Turnaround Time : 10.33
Average Response Time : 0.33

HOSPITAL PATIENT SCHEDULER

1. Non-Preemptive Priority
2. Preemptive Priority
3. Compare Both
4. Exit

```
suhani@siri:~/Hospital-Patient-Scheduler$ cd /home/suhani/Hospital-Patient-Scheduler
suhani@siri:~/Hospital-Patient-Scheduler$ gcc src/main.c src/scheduler.c src/fileio.c src/metrics.c
suhani@siri:~/Hospital-Patient-Scheduler$ ./a.out
```

HOSPITAL PATIENT SCHEDULER

1. Non-Preemptive Priority
2. Preemptive Priority
3. Compare Both
4. Exit

Enter choice: 1

NON-PREEMPTIVE PRIORITY

1. With Arrival Time
2. Without Arrival Time
3. Back to Main Menu

Enter choice: 2

NON-PREEMPTIVE WITHOUT ARRIVAL TIME

Enter number of processes (1-100): 3

----- Process 1 -----
Enter Process ID: 4
Enter Burst Time: 2
Enter Priority: 4

----- Process 2 -----
Enter Process ID: 6
Enter Burst Time: 8
Enter Priority: 4

----- Process 3 -----
Enter Process ID: 3
Enter Burst Time: 8
Enter Priority: 6

Aging Configuration

1. Disable Aging
2. Enable Aging

Enter choice: 2
Enter aging threshold: 4

Save results to output file?

1. No
2. Yes

Enter choice: 2
Enter output file name: f1

NON-PREEMPTIVE PRIORITY RESULTS

Process States

4 : Terminated
6 : Terminated
3 : Terminated

Execution Order

4 -> 6 -> 3

Gantt Chart

| 4 | 6 | 3 |
0 2 10 18

Process Results

ID	Arrival	Burst	Priority	Completion	Turnaround	Waiting	Response
4	0	2	4	2	2	0	0
6	0	8	4	10	10	2	2
3	0	8	6	18	18	10	10

Average Waiting Time : 4.00
Average Turnaround Time : 10.00
Average Response Time : 4.00

Results written to: f1

Performance Comparison

Non-Preemptive Priority		Preemptive Priority	
Average Waiting Time	1.67	Average Waiting Time	1.67
Average Turnaround Time	5.33	Average Turnaround Time	5.33
Average Response Time	1.67	Average Response Time	1.67

Non-Preemptive Gantt Chart

IDLE	P2	P1	P3
0 - 3	3 - 5	5 - 8	8 - 14

Preemptive Gantt Chart

IDLE	P2	P1	P3
0 - 3	3 - 5	5 - 8	8 - 14

Process Comparison

Process	NP Waiting	P Waiting	NP Turnaround	P Turnaround	NP Response	P Response
P1	0	0	3	3	0	0
P2	0	0	2	2	0	0
P3	5	5	11	11	5	5

Scheduling Metrics

Average Waiting Time	1.67	Average Treatment Turnaround	5.33	Average Response Time	1.67
----------------------	------	------------------------------	------	-----------------------	------

Patient Scheduling Results

Patient	Arrival	Treatment	Medical Priority	Treatment Completion	Turnaround Time	Waiting Time	Response Time
P1	5	3	1	8	3	0	0
P2	3	2	2	5	2	0	0
P3	3	6	2	14	11	5	5

Gantt Chart

IDLE 0 - 3	P2 3 - 5	P1 5 - 8	P3 8 - 14
---------------	-------------	-------------	--------------

Treatment Sequence: P2 → P1 → P3
No preemption detected.

Scheduling Metrics

Average Waiting Time	1.67	Average Treatment Turnaround	5.33	Average Response Time	1.67
----------------------	------	------------------------------	------	-----------------------	------

Patient Scheduling Results

Patient	Arrival	Treatment	Medical Priority	Treatment Completion	Turnaround Time	Waiting Time	Response Time
P1	5	3	1	8	3	0	0
P2	3	2	2	5	2	0	0
P3	3	6	2	14	11	5	5

Gantt Chart

IDLE	P2	P1	P3
0 - 3	3 - 5	5 - 8	8 - 14

Treatment Sequence: P2 → P1 → P3
No preemption detected.